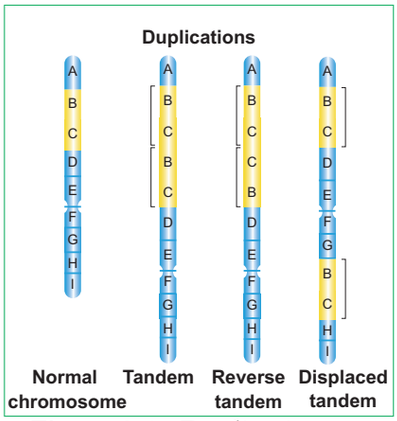


Figure 3.15 Duplication

3.Inversion

A rearrangement of order of genes in a chromosome by reversed by an angle 1800. Thisinvolve two chromosomal breaks and reunion.During this process there is neither gain nor lossbut the gene sequences is rearranged. Inversionwas first reported in ***Drosophila*** by Sturtevant(1926). There are two types of inversion,paracentric and pericentric (Figure 3.16).

i. Paracentric inversion: An inversion whichtakes place apart from the centromere.

ii. Pericentric inversion: An inversion thatincludes the centromere.

Inversions lead to evolution of a new species.

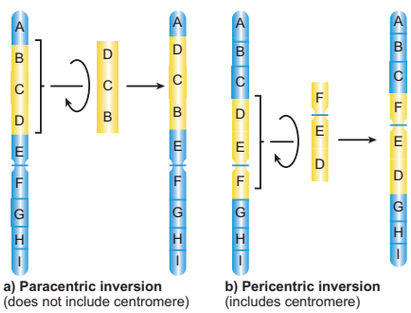


Figure: 3.16: Inversion

4. Translocation

The transfer of a segment of chromosome to a non-homologous chromosome is called translocation. Translocation should not be confused with crossing over, in which an exchange of genetic material between homologous chromosome takes place. Translocation occurs as a result of interchange of chromosome segments in non-homologous chromosomes. There are three types (Figure 3.17)

i. Simple translocation
ii. Shift translocation
iii. Reciprocal translocation

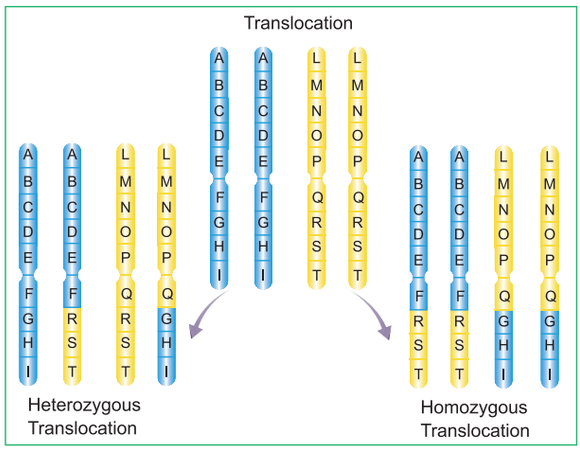


Figure: 3.17: Translocation

3.6 DNA Metabolism in Plants

As the repository of genetic information, DNA occupies a unique and central place among biological macromolecules. The structure of DNA is a marvelous device for the storage of genetic information. The term "DNA Metabolism" can be used to describe process by which copies of DNA molecules are made (replication) along with repair and recombination.

In this chapter we briefly discuss about the DNA metabolism in plants

DNA Replication: In the double helix the two parental strands of DNA separate and each parental strand synthesizes a new complementary strand. DNA replication is semiconservative, i.e each new DNA molecule conserves one original strand.

DNA Repair: How is genomic stability maintained in all living organisms? How do organisms on earth survive? What is essential for their survival?

DNA is unique because it is the only macromolecule where the repair system exists, which recognises

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